SPARCS Data Extract – Scope and Rationale (WP1)

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We download about 6 million adult stays from all 8 New York service areas for four years (2017, 2018, 2019, 2023), plus statewide frequency tables, instead of about 18 million rows in eight full yearly files. Everything is filtered on the server through the Health Data NY API by `download_sparcs.py` and saved as Parquet.

- **Years:** 2017 defines the frozen "typical stay" ruler and trains the model; 2018–2019 validate it; 2023 is the KPI baseline and the test year, and 2024 is added as a second test year when published. 2022 is not downloaded.
- **Health systems:** all 8 service areas at stay level, for training and for comparing systems. Finger Lakes (17 hospitals) remains the business-case system for the KPI and the cost scenario.
- **Rows:** adult acute medical-surgical stays only, about 1.48 million of 2.13 million statewide in 2023.
- **Columns:** 18 of 33, split into admission-known inputs, the target, and discharge-only analysis columns.
- **Status:** script written and tested offline against a mocked API; the filters were checked with live API counts. Run it once to produce the files.

How we built the extract: schema first, then filter at the API

We read each year's schema before downloading anything, then used it to write filters that the API applies on the server. Only the rows and columns the decision needs ever leave Health Data NY.

1. **Read the schema.** For each year, the API metadata (`/api/views/<dataset id>.json`) lists every column with its API name, type and null count. That is how we found that all 33–34 columns are text, that two columns were renamed after 2016, that CCS became CCSR, and that a null facility marks a redacted stay.
2. **Profile the values.** Small aggregate queries (stays by MDC, age group, admission type, discharge destination, diagnosis group) confirmed the exact codes the filters need, such as MDC `'14'` as two-digit text, the age band `'0 to 17'` and the destination `'Expired'`.
3. **Turn each decision into a query.** Each row rule becomes a `$where` clause, each column choice a `$select`, and the ruler and trend tables a `$group` count.

4. **Download only the result.** Results are paged 50,000 rows at a time and saved as Parquet, with the exact query, row count and download time recorded in `manifest.json`.

Why filter at the source rather than download full files:

- **Volume:** about 6 million stays and 18 columns instead of about 18 million rows and 33–34 columns, so it fits on a laptop.
- **Data minimisation:** we take only what the decision needs, a GDPR principle we apply even though the data is already de-identified.
- **Reproducible and auditable:** every rule is code in one settings block. Re-running the script gives the same extract, and the manifest shows exactly what was asked for.
- **Trade-off:** anything filtered out can't be re-analysed without a new download. Because the API remains the source of record, changing a rule means changing one setting and running again. In WP2 the landed extract is our raw layer: stored as the API returned it and never edited.

Which years and why

We download stay-level data for 2017, 2018, 2019 and 2023 only; the trend years enter only as counts. Each year has one job:

Year(s)	Role	What we download	Why this year
2017	Reference ("ruler") and model training	Statewide frequency table + statewide stays	A clean, pre-COVID year with about 1.5 million stays in scope: enough for stable P75s
2018–2019	Model validation	Statewide stays	Same pre-COVID regime: tuning is not distorted by the pandemic
2023	KPI baseline and model test	Statewide stays	Latest file published (2,125,754 stays); the brief asks for the KPI's current value. Testing here shows how the model copes with the post-COVID shift
2024	Second test year	Added when published	Not on Health Data NY as of today

Year(s)	Role	What we download	Why this year
2016–2021, 2023	Trend ("cost of doing nothing")	One aggregate table, no stay-level rows	Shows whether excess bed-days are rising

Years left out, and why:

- **Before 2016:** the US switched diagnosis coding from ICD-9 to ICD-10 on 1 October 2015, so 2015 is a mixed year and earlier years are not comparable.
- **2020–2021:** COVID changed who was admitted and how long they stayed. They appear in the trend, flagged, but never as reference, training or baseline.
- **2022:** team decision. With 2018–2019 for validation and 2023 for testing it has no role, and leaving it out keeps the extract smaller.

The ruler is **frozen for the life of the project**: typical stay = P75 of length of stay by APR-DRG × severity, New York State 2017. Groups with fewer than 30 stays in 2017 use the P75 of their whole diagnosis group. WP4 measures impact against the same ruler.

Which health systems and why

We keep stay-level data for **all 8 service areas**, about 1.5 million stays a year. Our business case is written for **a health system modelled on the Finger Lakes region** (Rochester, 17 hospitals, 137,867 stays in 2023): the KPI baseline, the cost scenario and the WP4 impact are reported for it, and the rest of the state is its benchmark.

Service area	Hospitals	Stays in scope, 2019	Role
New York City	—	688,644	Training, benchmark
Long Island	22	272,018	Training, benchmark
Hudson Valley	32	166,786	Training, benchmark
Capital/Adirondack	26	118,107	Training, benchmark
Finger Lakes	**17**	**112,136**	**Training + business-case system**
Western NY	24	111,040	Training, benchmark
Central NY	25	109,802	Training, benchmark
Southern Tier	4	22,649	Training, benchmark

"Stays in scope" = adults, excluding obstetrics, psychiatry and substance use (API count, 2019).

Why all 8 areas at stay level:

- **More training data:** about 15 times more stays than one area, so rare diagnosis × severity groups have enough cases.
- **A harder, fairer test:** the model can be checked on each area separately, including one it was not tuned on. That shows whether it generalises beyond one system.
- **Benchmarking:** the same diagnosis and severity can be compared across 8 areas and about 200 hospitals (hypothesis H3: the problem is a matter of process).

Why one business-case system rather than all hospitals:

- **The decision needs an owner.** Early discharge planning is decided inside one organisation, by its Director of Patient Flow and Head of Case Management. The state as a whole has no such person, and the approved proposal names a multi-hospital health system.
- **The money lands on one budget.** Case-manager costs and bed-day savings belong to one P&L; a statewide total adds up about 200 hospitals that share no budget.
- **WP4 must move the WP1 number.** Impact is measured against the baseline of the organisation that runs the programme.
- **Statewide still counts:** it sizes the opportunity ("if every New York hospital matched the best performers"), sets the benchmark and trains the model.

Why the Finger Lakes region:

- **Realistic size:** about 100,000 adult stays a year, the size of a real regional health system.
- **A real mix of hospitals:** one academic centre (Strong Memorial, 33,197 stays in 2019), three large city hospitals (Rochester General, Unity, Highland) and community hospitals.
- **Easy to change:** it is one setting (`SYSTEM_HSA`) in the script and a flag (`business_case_system`) in the data.

Which rows and why

We keep **adult acute medical-surgical stays**: the beds where discharge planning applies and where long stays block admissions. In 2023 that leaves about 1.48 million of 2,125,754 stays statewide, and about 99,800 of 137,867 in Finger Lakes. The rules below run in this order, in the API query itself, so excluded rows are never downloaded.

#	Rule	Why	Finger Lakes 2023, rows left
0	All stays	—	137,867
1	Drop stays with no facility (redacted)	Sensitive stays hidden at source; cannot be assigned to a hospital (5,333 statewide in 2023)	137,867
2	Drop newborns, obstetrics (MDC 15, 14), psychiatry and substance use (MDC 19, 20)	Separate wards and bed pools, with their own discharge processes	105,472
3	Drop under-18s	Paediatric beds are a separate pool	99,880
4	Drop rehabilitation (APR-DRG 860) and ungroupable codes (955, 956)	Rehab is a different bed type; ungroupable stays have no valid ruler	99,817
5	Flag, don't drop: deaths and transfers to another acute hospital	The stay did not end with a discharge anyone could plan. Out of the KPI, kept for the model	script reports

Rules for values inside the rows that are kept:

- **Length of stay `120 +`:** set to 120 and flagged (`los_capped`). These are real and expensive stays, so they stay in.
- **Missing or zero costs:** the row stays; it is left out only of cost calculations, so bed-days are not undercounted.
- **Thin groups:** a diagnosis × severity group with fewer than 30 reference stays uses the P75 of its whole diagnosis group instead.

The script writes the exact count after every rule, for every year, to `funnel.csv`. Rule 5 uses discharge information: fine for measuring the KPI, but the WP3 model must score every patient at admission.

Which columns and why

We keep 18 of the 33 columns. The test for each one: **would the hospital know it on the day of admission?** Columns known only at discharge are kept for analysis but can never be model inputs, because that would leak the outcome.

Group	Columns (API names)	Why
Known at admission (model inputs)	`discharge_year`, `hospital_service_area`, `permanent_facility_id`, `age_group`, `gender`, `zip_code_3_digits`, `type_of_admission`, `emergency_department_indicator`, `payment_typology_1`, `apr_mdc_code`, `apr_drg_code`, `apr_severity_of_illness_code`, `apr_medical_surgical`	Who the patient is, how they came in, who pays, what they have. Diagnosis group and severity define the ruler
Target	`length_of_stay`	The KPI and the model target
Known only at discharge (analysis only)	`total_costs`, `patient_disposition`	Cost per day for the cost scenario; discharge destination shows the root cause (waiting for a nursing home place)
Fairness check only	`race`, `ethnicity`	To test that the flag treats groups equally. Never model inputs

Dropped, and why:

Columns	Reason
`hospital_county`, `operating_certificate_number`	Duplicate the facility id
`facility_name` and all `*_description` columns	Moved to lookup tables; the facts keep only codes
`ccsr_diagnosis_*` / `ccs_diagnosis_*`	Duplicates the diagnosis group, and its classification changed between years
`ccsr_procedure_*` / `ccs_procedure_*`	Procedures happen during the stay: leakage
`apr_risk_of_mortality`	Coded at discharge; overlaps with severity

Columns	Reason
`payment_typology_2`, `payment_typology_3`	Secondary payers; the primary payer is enough
`birth_weight`, `abortion_edit_indicator`	Newborns and obstetrics are out of scope
`total_charges`	List prices, not costs; `total_costs` is the better figure

One honest caveat: diagnosis group and severity are **coded at discharge**. In production the model would use the provisional diagnosis known at admission, so in the EDA they stand in for it.

Column names changed between releases. The API names are the same from 2017 to 2023; in 2016 `health_service_area` and `facility_id` are the old names, and the script maps them.

The files and the API queries

`download_sparcs.py` builds three tables plus lookups, all filtered on the server. Only Table B holds stay-level rows; A and C are frequency tables (one row per diagnosis × severity × length of stay, with a count), which give exact P75s and KPIs at a fraction of the size.

File	Content	Years	Used for
`A_ruler_freq_statewide_2017.parquet`	Statewide count of stays per DRG × severity × LOS	2017	Building the ruler
`A_ruler_p75_2017.parquet` / `.csv`	The frozen P75 per DRG × severity, with thin-group fallback	2017	The KPI, the target flag
`B_stays_statewide_<year>.parquet`	One row per stay, all 8 areas, 18 columns + `los`, `los_capped`, `kpi_eligible`, `business_case_system`	2017, 2018, 2019, 2023	EDA, baseline, cost scenario, model (train / validate / test)

File	Content	Years	Used for
`C_trend_freq_2016_2023.parquet`	Count of stays per service area × DRG × severity × LOS	2016–2021, 2023	Trend line, benchmark
`funnel.csv`	Rows left after each rule, statewide and Finger Lakes	Each stay year	Data quality slide
`lookup_apr_drg.csv`, `lookup_apr_mdc.csv`, `lookup_facilities_<year>.csv`	Code → name	2023 / per year	Readable labels in charts
`manifest.json`	Settings, every API query sent, row counts, download time	—	Lineage (WP2)

To run it (Python 3.10+):

```
pip install requests pandas pyarrow
python download_sparcs.py                  # everything into ./data
python download_sparcs.py --only stays     # one table at a time
```

The stay query sent for each year (Table B), as an example:

```
GET https://health.data.ny.gov/resource/46xm-urtu.csv
  $select = discharge_year, hospital_service_area, permanent_facility_id, …
(18 columns), facility_name
  $where  = permanent_facility_id IS NOT NULL
            AND type_of_admission != 'Newborn' AND apr_mdc_code NOT IN
('14','15','19','20')
            AND age_group != '0 to 17'
            AND apr_drg_code NOT IN ('860','955','956')
  $order  = :id      $limit = 50000      $offset = 0, 50000, …
```

Dataset ids per year: 2016 `gnzp-ekau` · 2017 `22g3-z7e7` · 2018 `yjgt-tq93` · 2019 `4ny4-j5zv` · 2020 `nxi5-zj9x` · 2021 `tg3i-cinn` · 2023 `46xm-urtu` (2022 `5dtw-tffi` is listed in the script but not downloaded). Each statewide stay year is about 30 pages of 50,000 rows, so the full run takes roughly 30–60 minutes; a free app token (`--app-token`) avoids throttling.

Known defects and gaps

The extract can support the WP1 decision, but these limits must be stated on the data slide.

Defect or gap	Effect	What we do
No patient id	Readmissions can't be tracked per patient	Use the hospital-level readmission dataset (PPR rates, `amqp-cz9w`) to check that shorter stays don't raise readmissions
No dates, only the year	No seasonality, no weekend-admission effect	Listed as a gap; the hospital's admission system (ADT) has them in production (WP2)
Diagnosis group and severity coded at discharge	Slight optimism in the EDA	Stated as a proxy for the admission diagnosis
`120 +` top code on LOS	The longest stays are understated	Set to 120 and flagged
Costs are estimated from charges by each hospital	Cost per day is approximate	Cross-check with the Cost Transparency dataset (`7dtz-qxmr`); test 3 scenarios
Grouper and coding updates across years	Older trend years are slightly less comparable	Trend shown with a note; 2020–2021 flagged as COVID
Redacted facilities	About 5,000–9,500 stays a year can't be placed	Dropped and counted
No social factors (lives alone, carer, housing)	A known driver of delayed discharge is missing	Gap for WP2; 3-digit zip can be linked to census deprivation data
No live bed census or ward	The bed-demand forecast can't be validated here	Gap for WP2

Privacy: the files are de-identified at source by NYS DOH (age bands, 3-digit zip, no dates, small cells redacted). No consent is needed for reuse under the NYS DOH data use policy, with attribution.

Sources

- Hospital Inpatient Discharges (SPARCS De-Identified): 2023 — metadata
- 2019 metadata · 2018 · 2017 · 2016 · 2022
- SPARCS Cost Transparency — metadata
- Counts in this doc: live API queries on Health Data NY, run 25 October 2026